Tether Experiment: PocketQube Deployed from a CubeSat

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***Abstract*—This paper demonstrates the feasibility of tethering small satellite types such as PocketQube and CubeSats to achieve system stability via gravity gradient torque. By doing so, collisions between the satellite pair can be prevented through sufficient tension between them. A preliminary design of a CubeSat mission, where a PocketQube is deployed from a 3U CubeSat, is presented. This baseline design shows basic calculations of system torque and 25 m-long tether tension. RF and optical communication between the satellites and imaging of the CubeSat from the PocketQube are additional technology experiments on the proposed mission.**

Keywords - PocketQube, space tether, CubeSat, optical communication, gravity gradient

I. Introduction

A space tether is a long, strong cable or line that connects two objects in space and has the potential to offer numerous benefits for space missions. They can be made of various materials, including Kevlar, Spectra, and Zylan, which have high tensile strength and are lightweight. The tether is designed to withstand the harsh conditions of space, including extreme temperatures and radiation. The length of the tether can range from a few meters to several kilometers, depending on the intended use [1].

The concept of using tethers in space missions has been around for several decades. However, recent advances in materials science and space technology have made tether technology a viable option for a wide range of space missions. In this paper, a short overview of space tether technology, including its history, types, and applications, is provided in Section II. CubeSat is a type of small satellite that has revolutionized space exploration and utilization by offering affordable and accessible options for conducting space missions. CubeSats are standardized and designed to be minor, typically measuring 10 x 10 x 10 cm in size, and have a mass of no more than 1.33 kg. The CubeSat is intended to be a low-cost, standardized platform for conducting space missions, enabling universities, research institutions, and companies to develop and launch their own missions. The CubeSat is designed to be modular, with standard interfaces that allow for easy integration of different components, such as sensors, communication systems, and propulsion systems. The CubeSat's small size and low cost make it an attractive option for a wide range of applications, from testing new technologies in space to conducting scientific research on a limited budget. CubeSats are used for a variety of applications, including Earth observation, remote sensing, climate research, and space exploration. They are also used for technology demonstration missions, such as testing new propulsion systems or communication technologies [1][2].

The PocketQube is smaller than the CubeSat, typically weighing less than 1 kg and measuring no more than 10 cm on each side. It is designed to perform a wide range of tasks, including Earth observation, remote sensing, and communication. The PocketQube is equipped with a variety of sensors, such as cameras, spectrometers, and radio transmitters, that enable it to collect data and communicate with ground stations. It is compact size and low cost make it an ideal platform for research and education, allowing students and researchers to conduct experiments and gather data in space. The PocketQube is a promising tool for advancing space science and exploration.

This paper presents a mission using a tether as a connection between a CubeSat and a PocketQube, which is deployed from the CubeSat. To achieve the project's aim, the system will utilize a momentum exchange tether to provide propulsion between the PocketQube and CubeSat satellites. The tether will generate tension using the gravity gradient momentum, enabling improved propulsion and power capabilities for the PocketQube. Besides the control of the tethered satellite pair, this technology demonstration mission also tests the PocketQube deployer, wi-fi, and optical communication links between the pair and a camera on the PocketQube that takes videos of the deployment and tethered flight. The tether provides a physical connection between the two satellites, enabling precise control and positioning of the PocketQube camera relative to the CubeSat. This allows for the capture of high-quality photographs of the CubeSat in motion, with minimal distortion or interference. The use of tethers between PocketQube and CubeSat can further enhance the capabilities of these small satellites by enabling various scientific and technological capabilities, including improved propulsion, communication, and stability. Propulsion can be achieved through the use of electrodynamic tethers, which generate electricity using the Earth's magnetic field. The tether can also be used for communication between PocketQube and CubeSat. This can

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be particularly useful for scientific missions that require real-time data transfer.

Additionally, the tether can provide improved stability for PocketQube and CubeSat. Small satellites can experience significant vibrations and disturbances in space, which can affect their performance. The tether can act as a stabilizer, reducing vibrations and improving the overall stability of the system. Tethered formation flying can also enable concepts for bistatic radar (where one satellite is the transmitter while the other is the receiver) or interferometry. The inter-satellite communication links can be particularly useful for such missions.

II. Background

A. Overview of Space Tethers

In the 1960s, scientists introduced the concept of a space elevator - a lengthy tether capable of conveying individuals and freight from the Earth's surface to space. This is where the notion of employing tethers in space missions originated. While the concept of a space elevator remains a subject of active research and development, other types of space tether technology have been successfully demonstrated in space. Space tether technology has the potential to revolutionize space exploration and utilization by offering a range of benefits, including propulsion, energy harvesting, debris mitigation, formation flying, and reentry. The space tether might be used for various purposes, including research into tether propulsion, and tidal stabilization. The techniques are analyzed in 5 main techniques such as Momentum Exchange Tether, Electrodynamic Tether, and Deorbit Tether. Each type of space tether has unique properties and technical details that make it suitable for specific applications [14].

- Momentum Exchange Tether is a type of space tether that exchanges momentum between two objects in space. The tether is deployed from a spinning object, such as a spacecraft, and attached to a non-spinning object, such as a satellite. The momentum from the spinning object is transferred to the non-spinning object, causing it to move in a controlled manner. The tethers can be used for a variety of applications, including attitude control, formation flying, and as a means of transporting cargo between two objects in space [14].
- Electrodynamic Tether is a type of space tether that generates electricity using the Earth's magnetic field. The tether is deployed from a spacecraft and interacts with the Earth's magnetic field, creating a voltage potential. The voltage potential is then used to generate electricity, which can be used to power the spacecraft or other systems. The tethers can be used for a variety of applications, including energy harvesting, attitude control, and as a means of deorbiting satellites [14].
- Deorbit Tether is a type of space tether that is used to deorbit spacecraft or debris. The tether is deployed from a spacecraft or satellite and interacts with the Earth's atmosphere, creating drag that slows down the spacecraft or debris. The tether eventually burns up in the atmosphere, taking the spacecraft or debris with it. The tethers can be used for a variety of applications, including debris mitigation, satellite deorbiting, and as a means of spacecraft's disposing at the end of their mission [14].

B. Tether Theory

The tether system is a space mechanism used to maintain a constant distance between two objects. A connecting element such as a rope, cable, or wire with a predetermined length is created between the two objects. This connecting element is stretched by tension forces as it moves in space, thereby maintaining the distance between the objects that make up the system [4].

This paper describes a tether system used between a CubeSat and a PocketQube. The tether system can be created using a connecting element such as a cable or wire between the CubeSat and PocketQube. This connecting element is tensioned to maintain a constant distance between the CubeSat and PocketQube. The distance is determined by the length of the connecting element. The tether system is necessary to maintain the distance because the speed of the CubeSat in orbit varies due to variable gravitational force and other forces based on its position in orbit. This variability can cause deviation in the position observed by the PocketQube, which can lead to changes in the distance. The tether system is designed to correct for these deviations [5].

The process of setting up a space tether between two satellites involves several crucial steps. Firstly, it is essential to determine the appropriate type of space tether based on the specific needs and requirements of this mission. This may involve considering factors such as the desired length, strength, and durability of the tether.

Once the appropriate type of space tether has been identified, the next step is to design and construct the tether. This will require careful consideration of the materials and components needed to ensure the tether is strong and durable enough to withstand the harsh conditions of space.

After the tether has been designed and constructed, the next step is to deploy it. This may involve using a suitable method, such as a spring-loaded mechanism, motor, or pyrotechnic device, depending on the design of the satellite and tether. Once the tether has been deployed along with the PocketQube, the next step is to establish a connection between the two satellites. This may involve using radio communication, optical communication, or physical connectors, depending on the specific requirements of the mission. Finally, it is crucial to conduct tests to verify the functionality of the tether. This may involve measuring power and data transfer between the two satellites, as well as ensuring the tether is stable and secure.

Working with a space tether requires careful planning and execution, as the tether can be sensitive to changes in temperature, radiation, and other environmental factors. However, if used effectively, a space tether can provide a valuable tool for space exploration and technology development [6].

C. Tethered Missions

There are several realized space tethers missions;

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The Tethered Satellite System (TSS): In 1992, NASA launched the TSS, which consisted of a 20-kilometer-long tether that connected a satellite to the Space Shuttle. The mission aimed to study the dynamics of the tether in space, as well as to test the feasibility of using tethers for power generation and propulsion. However, the mission was cut short when the tether snapped [3].

The TiPS mission: In 1996, NASA launched the Tether Physics and Survivability (TiPS) mission, which aimed to study the behavior of a TSS in the harsh environment of space [3]. The mission deployed a 5-kilometer-long tether that connected two satellites and studied the effects of space debris impacts and other environmental factors on the tether.

The Orbital Debris Removal (ODR) mission: There are several ongoing missions that aim to use space tethers for orbital debris removal, such as the Remove Debris mission, which was launched in 2018. The mission uses a tethered net to capture and deorbit space debris and has successfully demonstrated its capability in multiple tests [13].

The Lunar Elevator mission: In 2020, a team of researchers from Japan proposed the concept of a "Lunar Elevator," which would use a space tether to transport cargo and humans from the surface of the Moon to a lunar space station in orbit. The mission would use a 100,000-kilometer-long tether that would be anchored to the lunar surface and would extend into space, allowing payloads to be lifted into orbit [3].

III. Proposed System

A. Overall System Design

The proposed platform is intended to operate in a low Earth orbit that is synchronized with the sun at an altitude of around 400 km. The mission is planned to last three months.

The design of the platform features a 1P PocketQube, which measures 5 x 5 x 5 cm, which is integrated into a 3U CubeSat and placed within a container. A 25 m-long tether is also deployed beneath the CubeSat and co-located with the PocketQube. The deployment system is enclosed within the container, which is equipped with rails, and the tether is held coiled around a spool. Once the CubeSat has completed its primary mission, the container- is opened, and the launch system is activated, causing the PocketQube to be ejected along with the tether. The tether takes approximately 5 minutes to extend fully and is then maintained in a perpendicular position to the orbit. Its primary function is to sustain the distance and orientation between the CubeSat and PocketQube.

The PocketQube is equipped with a camera, which will enable the PocketQube to capture the CubeSat's video, an accomplishment that has not previously been achieved, and to record a video of the deployment. The communication payload of the PocketQube comprises two distinct communication networks: Wi-Fi and optical communication. The data collected by the PocketQube is transmitted to the CubeSat satellite receives data collected through these communication networks and relays it to the ground station.

The deployment system of the platform incorporates a rail system that ejects the PocketQube and tether from the container. Upon extension, the tether is maintained in a perpendicular position to the orbit [7].

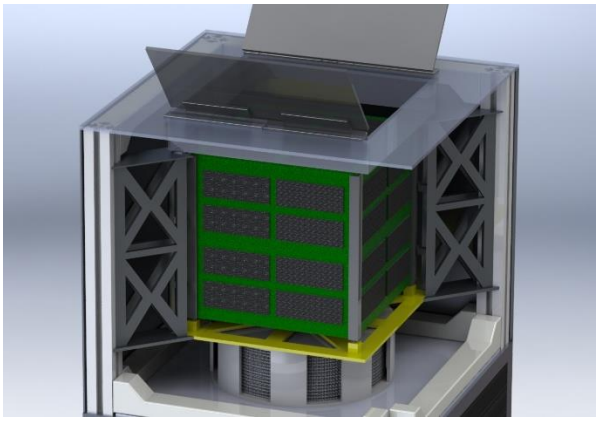


Fig. 1. Overall system design

B. PocketQube System Design

The PocketQube platform, measuring 5 cm on each side and weighing 550 gr (with a 10% design margin), comprises four interconnected subsystems. These subsystems include the electrical power subsystem, the mission computer, the communication system, and the attitude control system, each with unique features and capabilities.

The electrical power subsystem is designed to convert energy from the solar panel into the necessary power supply for the satellite. It includes a range of components, such as a maximum power point tracker (MPPT) charge integrator, a monitor to control current, a battery for energy storage, and a battery integrator to monitor the charge-discharge status of the battery. This subsystem provides the necessary power supply for the PocketQube while ensuring efficient usage. The solar panel cells of the PocketQube system cover all four surfaces and have an area of 81 cm², with a production capacity of 2.95 Watts [8][9]. Peak power consumption in each mode is shown in Table 1.

TABLE I. Peak Power in Main Modes

Modes	Peak Power (W)
Early Orbit Phase	0.85
Orbit Control Mod	1.1
Daytime (Full Sun)	1.946
Nighttime (Eclipse)	1.1
Safe Mode	2.49

The mission computer, on the other hand, is an electronic card that comprises a low-power microcontroller, a safety timer integrator, and a voltage regulator. The data is stored in MRAM memory to minimize data loss. It includes indicators that display different statuses for instant control and error analysis of the system. This enables external control of the system status, allowing for efficient management of the subsystems.

The Wi-Fi communication system, which uses the Lora WAN protocol, is designed to connect battery-powered objects and assets to the internet in regional, national, or global networks. This protocol meets the fundamental requirements of the Internet of Things (IoT), such as

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bidirectional communication, end-to-end security, mobility, and localization services.

The optical communication system for PocketQube utilizes LEDs as transmitters. This system enables high-speed data transfer between the satellite and Earth, while minimizing energy requirements through low power consumption.

The payload of the PocketQube comprises a camera and 5 LEDs. The camera is capable of capturing images and videos of the CubeSat at a resolution of 1.2 cm at a distance of 25 m, which are transmitted to the CubeSat using any of the links. Finally, the attitude control system utilizes magnetorquers to maintain the coarse attitude of the spacecraft during deployment.

C. *Cubesat System Design*

A CubeSat with a volume of 3U has been adopted as the baseline, 1U of which is allocated to the PocketQube deployer and related electronics. To maintain stability in Earth's orbit, a 3-axis stabilization system using reaction wheels is employed.

D. *PocketQube Deployment System*

The payload has been designed to be compatible with CubeSat of varying sizes, measuring 10 x 10 x 10 cm. It is housed within a compartment that contains a stationary PocketQube, which serves to maintain the position of the payload inside the CubeSat. The compartment also features a reel system to which the tether is attached, a cover system for pre-launch protection, and a rail-spring separation mechanism for launch. This separation mechanism comprises four rails made of anodized aluminum and four springs made of stainless steel with a high elastic modulus. The rails are designed with a low coefficient of friction to ensure smooth movement of the PocketQube during the launch. The friction coefficient is calculated based on the material properties and the expected operating conditions in space.

The friction force acting on the rails during deployment can be estimated to be approximately 0.0028 N.

Upon reaching the magnetic pole, the cover opens, initiating the launch process, this region is used because the magnetic field is perpendicular. The platform cover is held in place by a string until reaching the launch point, at which point the cover opens, and the launch mechanism engages, creating tension in the spring and releasing the PocketQube. The rails ensure a smooth exit for the PocketQube, and the tether is gradually released to maintain tension in accordance with the tether calculations, thus keeping the PocketQube at a stable distance from the CubeSat.

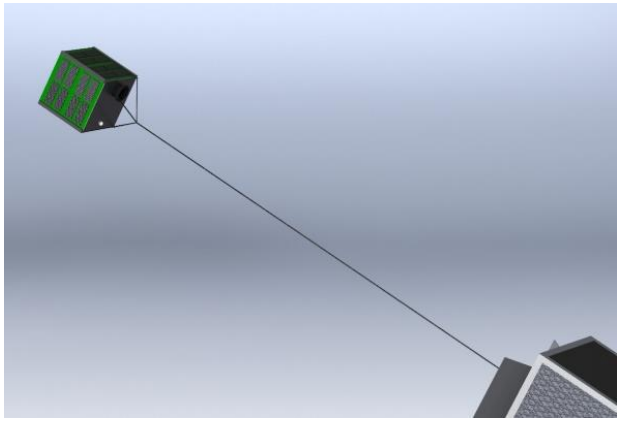


Fig. 2. PocketQube deployment system with tether

E. *Stability of the System*

The stability of the CubeSat and PocketQube systems connected with the tether is affected by various forces. These forces are generated by the movement and orbit of the satellites, thermal effects, gravity, electromagnetic effects, and other factors. The tether between two satellites is subjected to tension force [18]. This force depends on the length of the tether and the mass of the satellites. The longer the tether, the higher the tension force. Additionally, the movements of the satellites can increase or decrease the force. The tension force between two satellites on a Tether is calculated to be 0.000016 N.

Atmospheric effects are another factor that affects the motion of the satellite system. Atmospheric drag can cause the satellite system to drift in its orbit. Gravity forces are another factor that affects system stability. The distance between the two satellites and the length of the tether determines the intensity of gravitational effects. Electromagnetic effects are also a factor that affects system stability. Magnetic fields and other electromagnetic forces in space can affect the movements of the satellites and cause the tether to stretch.

In order to stabilize the system, the solar pressure torque affecting the satellite is minimized by using white coatings on the external surface of the satellite. The magnetic torque is also minimized by designing the system with materials that have less impact on the magnetic field used in the system. As a result of the calculations presented in this study, the gravity gradient value is found to be greater than all disturbance forces. Consequently, the system is stabilized with passive forces along the tension in the tether and at a distance equal to the length of the tether from the PocketQube to the CubeSat.

Optimization was carried out by adjusting the masses and tether length ratios of two satellites in order to reach the required gravity gradient torque for stabilizing the system [15].

The PocketQube TSS is designed to enhance the stability and pointing of the system under all disturbing and perturbing forces [18]. Although the PocketQube does not require active control and its direction can be controlled solely by the tension in the tether, it has a magnetic attitude control system. The current force calculations caused by disturbance torques

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on the satellite system in LEO are indicated according to their magnitudes of effect [2][14].

- Aerodynamic Torque
- Magnetic Torque
- Gravity Gradient Torque
- Solar Pressure Torque

Sample calculations shown in this document are for an orbit of 400 km, considering mass, center of gravity, aerodynamic pressure center, and cross-sectional area.

1) Aerodynamic Torque

Aerodynamic torque affects satellites in space is the rotational force generated by the interaction of the residual atmosphere and solar radiation pressure with the spacecraft's surface area and geometry, which can affect its attitude and orbital dynamics [17].

$$T_a = F \cdot (C_p - C_g) \quad (1)$$
$$F = 0.5 \cdot [\rho C_d \, AV^2] \quad (2)$$

Where T_a results from aerodynamic forces causing rotational motion on a satellite. F is the force caused by the airflow on the satellite. ρ is atmospheric density. A is the cross-sectional area of the satellite. However, the drag coefficient (C_d) is a coefficient used to measure the resistance of a satellite to airflow. Center of gravity (C_g) is the point where the weight force of a satellite is applied. Aerodynamic pressure center (C_p) is the point where the effect of the aerodynamic pressure distribution on a satellite is located [17].

$C_{d,PocketQube}$, $C_{d,CubeSat}$, $A_{PocketQube}$, $A_{CubeSat}$, $V_{circular,PocketQube}$, $V_{circular,CubeSat}$ are calculated as 2.499, 2.503, 0.0025 m^2, 0.03 m^2, 7668.617 m/s, 7668.631 m/s respectively. In addition, $C_p - C_g = 8.91\ m$.

Therefore, from the equation:

$$F_p = 0.5\left[(3.89 \cdot 10^{-12}) \cdot (2.499) \cdot (0.0025 m^2) \cdot \left(7668.617 \frac{m}{s}\right)^2\right] = 2.278 \cdot 10^{-8} N$$

$$F_c = 0.5\left[(3.89 \cdot 10^{-12}) \cdot (2.503) \cdot (0.03 m^2) \cdot \left(7668.631 \frac{m}{s}\right)^2\right] = 4.557 \cdot 10^{-8} N$$

2) Gravity Gradient Torque

The gravity gradient torque is a stabilizing force experienced by a satellite in space due to the variation in the gravitational field along its length, which can induce a torque, causing attitude oscillations or instability [15]. The gravity gradient torque value of the tether is calculated for PocketQube.

$$T_{g,total} = \frac{3 \cdot G \cdot M}{2 \cdot R^3} |I_y - I_z| \sin(2 \cdot (\theta)) \quad (3)$$

Where T_g is the Gravity Gradient Torque, the torque value resulting from the gradient in gravitational forces acting on different points of the satellite. R is the radius length of the orbit. I_z is the moment of inertia value in the z-axis of the system. I_y is the moment of inertia value on the y-axis of the system. G is the gravitational constant ($6.673 \cdot 10^{-11}$), and M is the mass of the Earth ($5.979 \cdot 10^{24} kg$). Theta (θ) is the angle between the system's Zenith and the z-axis [15].

$$T_{g,total} = \frac{3 \cdot 6.673 \cdot 10^{-11} \cdot 5.979 \cdot 10^{24} kg}{2 \cdot (6378 + 400\ km)^3} |302.23 - 0.021\ kg \cdot m^2| \sin(2 \cdot (5°))$$

$$T_{g,total} = 1.008 \cdot 10^{-4}\ N.m$$

3) Solar Pressure Torque

Solar pressure is a force exerted on a satellite in space due to the momentum transfer from photons in solar radiation, which can cause orbital perturbations and affect the satellite's attitude control [16].

$$T_{sp} = F(C_{ps} - C_g) \quad (4)$$
$$F = \frac{F_s}{C} A_s (1 + q) \cos\alpha \quad (5)$$

The important variables for solar-powered spacecraft include the surface area of the spacecraft (A_s) in square meters, the angle of incidence to the Sun (α) in degrees, and the surface reflectance (q). In addition, the center of solar pressure (C_{ps}) is a critical consideration. F_s is the solar constant ($1367 W/m^2$).The solar pressure of a PocketQube (F_p) and CubeSat (F_c) are expressed in Newtons, while the solar pressure of PocketQube ($T_{sp,p}$) and CubeSat ($T_{sp,c}$) refer to the Solar Pressure in general [16].

$$F_p = \frac{1367}{3 \cdot 10^8} \cdot (2.5 \cdot 10^{-3}) \cdot (1 + 1) \cdot \cos(45°) = 1.61102 \cdot 10^{-8}\ N$$
$$F_c = \frac{1367}{3 \cdot 10^8} \cdot (0.03) \cdot (1 + 0) \cdot \cos(45°) = 3.22205 \cdot 10^{-8}\ N$$

$$T_{sp,p} = (1.61102 \cdot 10^{-8})(0.0177) = 2.85151 \cdot 10^{-10}\ N.m$$
$$T_{sp,c} = (3.22205 \cdot 10^{-8})(0.0375) = 1.20827 \cdot 10^{-9}\ N.m$$

The solar force values in the system can be minimized by adjusting the surface reflectance and center of gravity values. The solar torque is neglected in the system since its effect is very small compared to the gravity gradient torque.

4) Magnetic Torque

The magnetic torque on each satellite can be minimized by using non-magnetic materials. Additionally, torques on each satellite can be minimized actively. There is no magnetic torque on the overall system.

System stability is ensured using gravity gradient torque. There are variables that can be used to minimize magnetic torque and drag torque. Since the solar pressure torque value is smaller than the gravity gradient value, it does not affect the system's stability.

F. Optical Communication Payload

Optical communication generally refers to the technology and systems that transmit information using light as the medium, typically through optical fibers. This method of communication is used to transmit data from one point to

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another using optical signals. Wi-Fi and optical communication are used to provide bidirectional communication between CubeSat and PocketQube. Additionally, Error Detection and Correction (EDAC) increases communication reliability. The target communication rate is 921.600 bps, which provides sufficient speed for the serial transmission of data. Programmable overhead rates are 16, and the time it takes for a photo to arrive is 15 seconds. In this way, reliable communication is established between CubeSat and PocketQube, enabling data and information exchange [11].

G. Camera Payload

The MQ013CG-ON with the Fujinon HF10HA-1B camera lens, which operates via a Universal Asynchronous Receiver/Transmitter (UART) interface, captures images in response to signals received from the CubeSat. Positioned flush with the tether and situated on the bottom of the PocketQube, the MQ013CG-ON (1280x1024) pixel resolution captures high-quality images from a distance of approximately 14 - 25 meters.

In addition to its imaging capabilities, the PocketQube also boasts Wi-Fi communication capabilities, further enhancing its data transmission capacity. Overall, the PocketQube represents an innovative and effective solution to the ever-growing demand for advanced space technology infrastructure and is poised to play a critical role in shaping the future of space technology [12].

IV. Conclusion

The feasibility of tethering small satellites such as PocketQube and CubeSats together to achieve system stability using gravity gradient torque has been demonstrated. This approach enables the satellite pair to prevent collisions by utilizing sufficient tension between them. Additionally, the mission design has shown that it is feasible to launch a PocketQube form a 3U CubeSat and perform technology experiments using this system. The team will continue to develop the mission further.

A study has shown the feasibility of connecting small satellites such as PocketQubes and CubeSats in a cube-shaped formation to achieve system stability using the gravity gradient torque. This approach allows the satellites to be tethered to each other using sufficient tension to prevent collisions. The objective of the study was to demonstrate the feasibility of tether stabilization under existing forces. The results of the stabilization data support the findings of previous studies on tethered satellite systems. Additionally, the study showed that it is possible to design a mission to launch a PocketQube from a 3U CubeSat and perform technology experiments using this system. Further analysis will be conducted in the study, and a communication concept will be developed.

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